

MATHEMATICS 1 MARKING GUIDE

1. (a) Symbolize and hence draw an electrical network

“It is not true that Rama studies Economics. But Rama studies Mathematics or Geography”.

$p \equiv \text{Rama studies economics}$

$q \equiv \text{Rama studies Mathematics}$

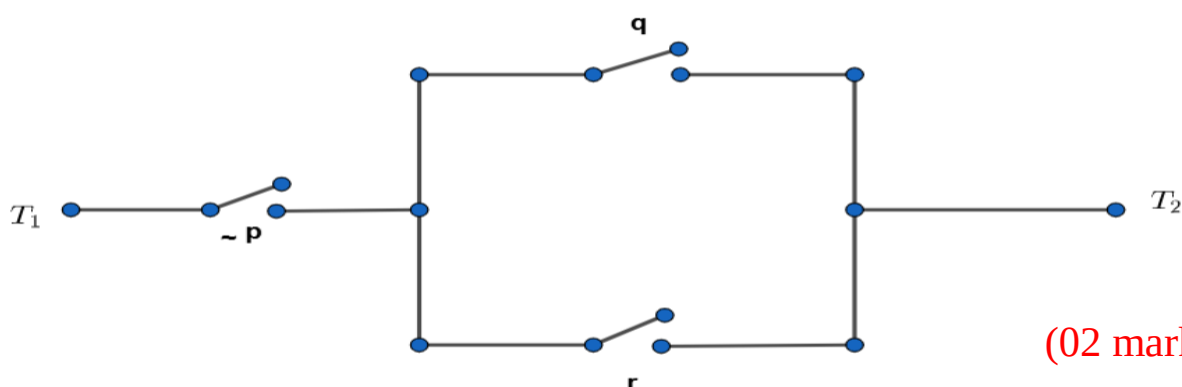
$r \equiv \text{Rama studies geography}$

then $\sim p \wedge (q \vee r)$

It electrical network

(01 marks)

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(02 marks)

- (b) Use truth table to check if $p \rightarrow (q \wedge r) \equiv (p \rightarrow q) \wedge (p \rightarrow r)$.

$$p \rightarrow (q \wedge r) \equiv (p \rightarrow q) \wedge (p \rightarrow r)$$

let $q \wedge r \equiv a$, $p \rightarrow q \equiv b$ and $p \rightarrow r \equiv c$

p	q	r	$q \wedge r$	$p \rightarrow a$	b	c	$b \wedge c$
T	T	T	T	T	T	T	T
T	T	F	F	F	T	F	F
T	F	T	F	F	F	T	F
T	F	F	F	F	F	F	F
F	T	T	T	T	T	T	T
F	T	F	F	T	T	T	T
F	F	T	F	T	T	T	T
F	F	F	F	T	T	T	T

(04 marks)

From the above $p \rightarrow a \equiv b \wedge c$

- (c) Use the laws of algebra of proposition to simplify $[(p \rightarrow q) \wedge \neg q] \rightarrow \neg p$

$$\begin{aligned}
 & [(p \rightarrow q) \wedge \sim q] \rightarrow \sim p && \text{Given} \\
 \equiv & \sim [(\sim p \vee q) \wedge \sim q] \vee \sim p && \text{Conditional} \\
 \equiv & [(\sim p \wedge \sim q) \vee q] \vee \sim p && \text{De - Morgan's} \\
 \equiv & [(\sim p \vee q) \wedge (\sim q \vee q)] \vee \sim p && \text{Distributive} \\
 \equiv & [(\sim p \vee q) \wedge T] \vee \sim p && \text{Negation} \\
 \equiv & (\sim p \vee q) \vee \sim p && \text{Identity} \\
 \equiv & (q \vee p) \vee \sim p && \text{Commutative} \\
 \equiv & q \vee (p \vee \sim p) && \text{Associative} \\
 \equiv & q \vee T && \text{Negation} \\
 \equiv & T && \text{Identity} \\
 \therefore & [(p \rightarrow q) \wedge \sim q] \rightarrow \sim p \equiv T && \text{(03 marks)}
 \end{aligned}$$

2. (a) Solve for x given $\operatorname{arc} \cosh 2x = \operatorname{arc} \sinh x$

$$\operatorname{arc} \cosh 2x = \operatorname{arc} \sinh x$$

$$\cosh^{-1} 2x = \sinh^{-1} x$$

$$2x = \cosh(\sinh^{-1} x)$$

$$2x = \sqrt{1 + \sinh^2 \sinh^{-1} x} \quad \text{(01 mark)}$$

$$2x = \sqrt{1 + x^2}$$

$$4x^2 = 1 + x^2$$

$$4x^2 - x^2 = 1 \quad \text{(01 mark)}$$

$$3x^2 = 1$$

$$x^2 = \frac{1}{3}$$

$$x = \pm \sqrt{\frac{1}{3}} = \pm \frac{\sqrt{3}}{3} \quad \text{(01 mark)}$$

(b) If $f(x) = \sinh x \cos x + \sin x \cosh x$,

(i) Find a simplified expression of $f'(x)$

$$f(x) = \sinh x \cos x + \sin x \cosh x$$

$$f'(x) = \left[\sinh x \frac{d}{dx}(\cos x) + \cos x \frac{d}{dx}(\sinh x) \right] + \left[\sin x \frac{d}{dx}(\cosh x) + \cosh x \frac{d}{dx}(\sin x) \right] \quad (01 \text{ marks})$$

$$f'(x) = \sinh x(-\sin x) + \cos x \cosh x + \sin x \sinh x + \cosh x \cos x \quad (01 \text{ marks})$$

$$f'(x) = \cos x \cosh x - \cancel{\sin x \sinh x} + \cancel{\sin x \sinh x} + \cosh x \cos x$$

$$f'(x) = 2 \cosh x \cos x \quad (01 \text{ marks})$$

(ii) Use the results in (b) (i) to evaluate

$$\int \frac{2}{\tanh x + \tan x} dx$$

$$\int \frac{2}{\tanh x + \tan x} dx$$

$$= \int \frac{2}{\frac{\sinh x}{\cosh x} + \frac{\sin x}{\cos x}} dx \quad \left(00 \frac{1}{2} \text{ marks} \right)$$

$$= \int \frac{2}{\frac{\sinh x \cosh x + \cosh x \sin x}{\cosh x \cos x}} dx \quad \left(00 \frac{1}{2} \text{ marks} \right)$$

$$= \int \frac{2 \cosh x \cos x}{\sinh x \cosh x + \cosh x \sin x} dx \quad (01 \text{ marks})$$

let $u = \sinh x \cos x + \cosh x \sin x$

$$\frac{du}{dx} = 2 \cosh x \cos x$$

$$dx = \frac{du}{2 \cosh x \cos x} \quad \left(00\frac{1}{2} \text{ marks}\right)$$

$$= \int \frac{\cancel{2 \cosh x \cos x}}{u} \frac{du}{\cancel{2 \cosh x \cos x}} \quad \left(00\frac{1}{2} \text{ marks}\right)$$

$$= \int \frac{du}{u}$$

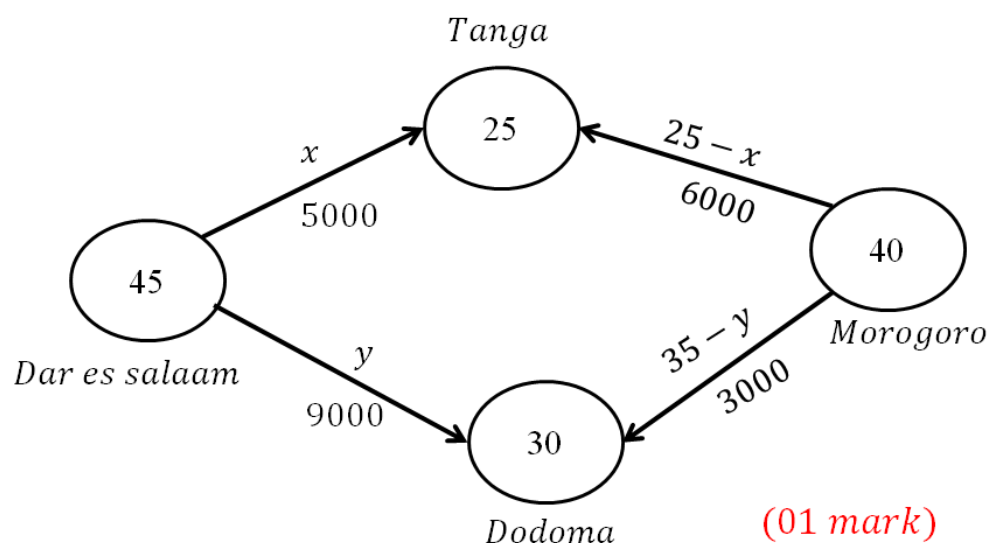
$$= \ln u + C \quad \left(00\frac{1}{2} \text{ marks}\right)$$

$$= \ln(\sinh x \cos x + \cosh x \sin x) + C$$

$$\therefore I = \ln(\sinh x \cos x + \cosh x \sin x) + C \quad \left(00\frac{1}{2} \text{ marks}\right)$$

3. A TV dealer has stores in Dar es Salaam and Morogoro and retailers in Tanga and Dodoma. The stores have a stock of 45 TV and 40 TV sets respectively while the requirements of the retailers are 25 and 30 TV sets respectively. If the cost of transporting a TV from Dar es Salaam to Tanga is Tsh. 5,000/= and from Dar es Salaam to Dodoma is Tsh. 9,000/=, from Morogoro to Dodoma is Tsh. 3,000/= and Morogoro to Tanga is Tsh. 6,000/=

- (a) How should the TV dealer supply the requested TV sets at minimum cost?



(01 mark)

Objective function

$$f(x, y) = 5000x + 9000y + 6000(25 - x) + 3000(30 - y)$$

$$f(x, y) = 1000(5x + 9y + 6(25 - x) + 3(30 - y))$$

$$f(x, y) = 1000(5x + 9y + 150 - 6x + 90 - 3y)$$

$$f(x, y) = 1000(-x + 6y + 240)$$

$$f(x, y) = 240,000 - 1000x + 6000y \quad (01 \text{ marks})$$

Inequalities

$$x \leq 25$$

$$y \leq 30 \dots\dots\dots \left(00\frac{1}{2} \text{ marks} \right)$$

$$x + y \leq 45 \dots\dots\dots \left(00\frac{1}{2} \text{ marks} \right)$$

$$25 - x + 30 - y \leq 40$$

$$55 - x - y \leq 40$$

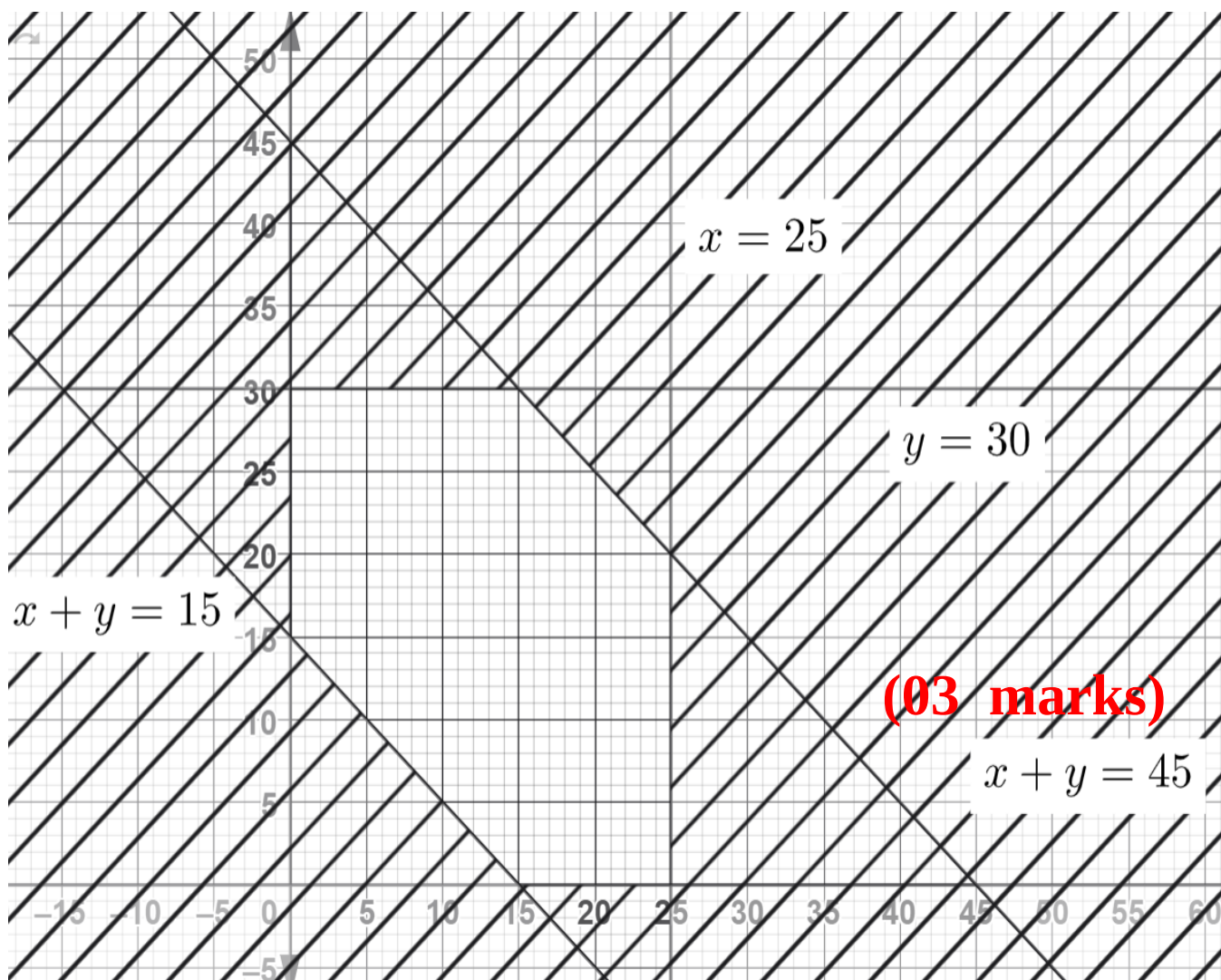
$$-x - y \leq -15$$

$$x + y \geq 15 \dots\dots\dots \left(00\frac{1}{2} \text{ marks} \right)$$

Non inequalities

$$x \geq 0$$

$$x \geq 0 \dots\dots\dots \left(00\frac{1}{2} \text{ marks} \right)$$



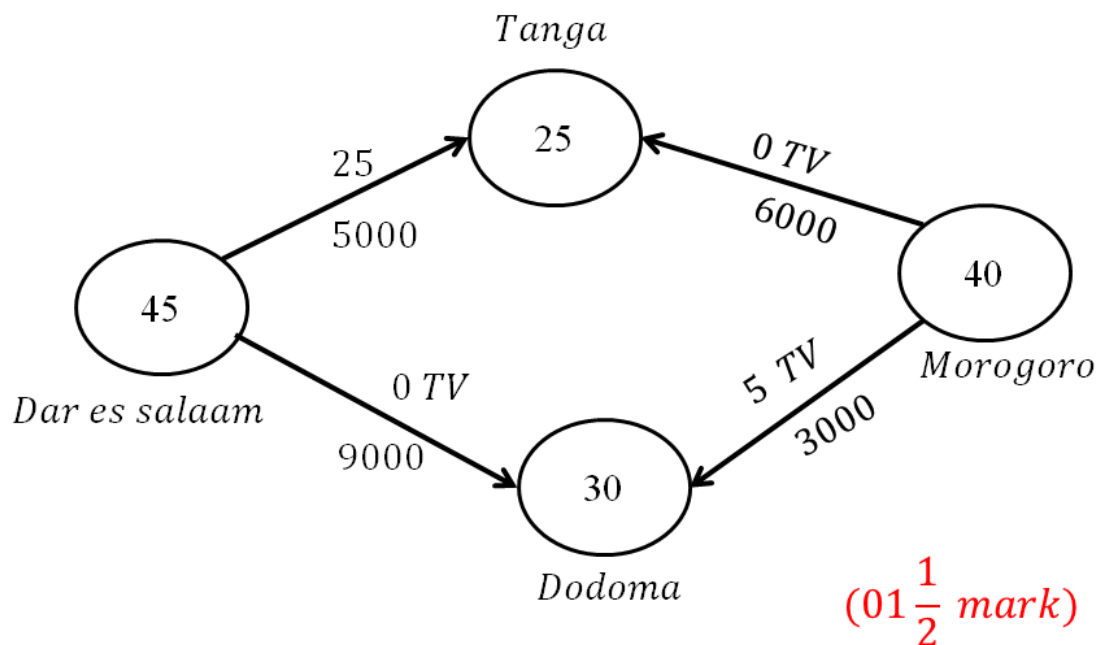
(03 marks)

CORNER POINT	$f = 240,000 - 1000x + 6000y$
A(0, 30)	420,000.00
B(0, 15)	330,000.00
C(15, 0)	225,000.00
D(25, 0)	215,000.00
E(25, 20)	335,000.00
F(15, 30)	405,000.00

(01 $\frac{1}{2}$ marks)

The minimum point is (25, 0)

(00 $\frac{1}{2}$ marks)



(b) The minimum cost is 215, 000/=

(00 $\frac{1}{2}$ marks)

4. (a) If for a distribution $\sum (x - 5) = 3$, $\sum (x - 5)^2 = 43$ and the total number of item is 18, find the mean and standard deviation.

$$\sum (x - 5) = 3 \text{ and } \sum (x - 5)^2 = 43, n = 18$$

$$\sum_{1}^{18} (x - 5) = 3$$

$$\sum_{1}^{18} x - \sum_{1}^{18} 5 = 3$$

$$\sum_{1}^{18} x - 5 \times 18 = 3$$

$$\sum_{1}^{18} x - 90 = 3$$

(00 $\frac{1}{2}$ marks)

$$\sum_{1}^{18} x = 90 + 3$$

$$\sum_{1}^{18} x = 93$$

(00 $\frac{1}{2}$ marks)

$$\sum (x - 5)^2 = 43$$

$$\sum_1^{18} (x - 5)^2 = 43$$

$$\sum_1^{18} (x^2 - 10x + 25) = 43$$

$$\sum_1^{18} x^2 - 10 \sum_1^{18} x + 25 \sum_1^{18} 1 = 43$$

$\left(00\frac{1}{2} \text{ marks} \right)$

$$\sum_1^{18} x^2 - 10 \times 93 + 25 \times 18 = 43$$

$$\sum_1^{18} x^2 - 930 + 450 = 43$$

$$\sum_1^{18} x^2 - 480 = 43$$

$$\sum_1^{18} x^2 = 523$$

$$\text{Mean} = \frac{\sum x}{n} = \frac{93}{18} = 5.17$$

$\left(00\frac{1}{2} \text{ marks} \right)$

$$\sigma^2 = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n} \right)^2$$

$$\sigma^2 = \frac{523}{18} - \left(\frac{93}{18} \right)^2$$

$\left(00\frac{1}{2} \text{ marks} \right)$

$$\sigma^2 = \frac{215}{9}$$

$$\sigma = \sqrt{\frac{215}{9}}$$

$$\sigma = 4.89$$

$\left(00\frac{1}{2} \text{ marks} \right)$

- (b) The mean weight of 150 students in a certain class is 60 kg. The mean weight of boys is 70 kg and that of girls in the class is 55 kg. Find the number of boys and girls in the class.

$$\text{Number of boys} + \text{Number of girls} = 150$$

$$n_b + n_g = 150$$

$$\bar{x} = \frac{\sum x_b + \sum x_g}{n_b + n_g} \quad \left(00\frac{1}{2} \text{ marks} \right)$$

$$\sum x_b = n_b \bar{x}_b = 70n_b$$

$$\sum x_g = n_g \bar{x}_g = 55n_g$$

$$60 = \frac{70n_b + 55n_g}{n_b + n_g} \quad \left(00\frac{1}{2} \text{ marks} \right)$$

$$60(n_b + n_g) = 70n_b + 55n_g$$

$$60n_b + 60n_g = 70n_b + 55n_g \quad \left(00\frac{1}{2} \text{ marks} \right)$$

$$10n_b - 5n_g = 0$$

$$\begin{cases} n_b + n_g = 150 \\ 10n_b - 5n_g = 0 \end{cases}$$

$$n_b = 50, \quad n_g = 100$$

$$\text{Number of boys is } 50$$

$$\text{Number of girls is } 100 \quad \left(00\frac{1}{2} \text{ marks} \right)$$

(c) The frequency distribution

x	p	$2p$	$3p$	$4p$	$5p$	$6p$
f	2	1	1	1	1	1

Where p is a positive integer, as a variance of 160. Find the value of p .

x	f	fx	x^2	fx^2
p	2	$2p$	p^2	$2p^2$
$2p$	1	$2p$	$4p^2$	$4p^2$
$3p$	1	$3p$	$9p^2$	$9p^2$
$4p$	1	$4p$	$16p^2$	$16p^2$
$5p$	1	$5p$	$25p^2$	$25p^2$
$6p$	1	$6p$	$36p^2$	$36p^2$
Σ	7	$22p$		$92p^2$

(02 marks)

$$\sigma^2 = \frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N} \right)^2$$

$$\sigma^2 = \frac{92p^2}{7} - \left(\frac{22p}{7} \right)^2$$

$$\sigma^2 = \frac{92p^2}{7} - \frac{484p^2}{49}$$

$$160 = \frac{644p^2 - 484p^2}{49}$$

(01 marks)

$$7840 = 160p^2$$

$$160p^2 = 7840$$

$$p^2 = \frac{7840}{160} = 49$$

$$p = \sqrt{49} = 7$$

(01 marks)

5. (a) Use laws of algebra to simplify $(A \cup B) \Delta B'$

$$\begin{aligned}
 & (A \cup B) \Delta B' && \text{Given} \\
 & \equiv ((A \cup B) - B') \cup (B' - (A \cup B)) && \text{Symmetric} \\
 & \equiv \left((A \cup B) \cap (B')' \right) \cup \left(B' \cap (A \cup B)' \right) && \text{Difference} \\
 & \equiv \left((A \cup B) \cap B \right) \cup \left(B' \cap (A \cup B)' \right) && \text{Complement} \\
 & \equiv \left((A \cup B) \cap B \right) \cup \left(B' \cap (A' \cap B') \right) && \text{De - Morgan's} \quad (01\text{mark}) \\
 & \equiv B \cup (B' \cap (A' \cap B')) && \text{Absorption} \\
 & \equiv B \cup (B' \cap (B' \cap A')) && \text{Commutative} \\
 & \equiv B \cup ((B' \cap B') \cap A') && \text{Associative} \\
 & \equiv B \cup (B' \cap A') && \text{Idempotent} \quad (01\text{mark}) \\
 & \equiv (B \cup B') \cap (B \cup A') && \text{Distributive} \\
 & \equiv \mu \cap (B \cup A') && \text{Complement} \\
 & \equiv (B \cup A') && \text{Identity} \\
 & \equiv (A' \cup B) && \text{Commutative} \quad (01\text{mark}) \\
 & (A \cup B) \Delta B' \equiv (A' \cup B)
 \end{aligned}$$

(b) Given $n(A \cap B) = 5$, $n(A \cup B) = 16$, $n(A \cup B)' = 4$ and $n(A - B) = 8$
find (i) $n(A \cup B')$

$$n(A \cap B) = 5$$

$$n(A \cup B) = 16$$

$$n(A \cup B)' = 4$$

$$n(A - B) = 8$$

$$n(A \cup B) = n(A - B) + n(B - A) + n(A \cap B)$$

$$16 = 8 + n(B - A) + 5$$

$$n(B - A) = 16 - 13 = 3$$

$$n(B - A) = n(B) \text{ only} = 3$$

$$n(A - B) = n(A) \text{ only} = 8$$

$$n(A \cup B') = n(A) + n(B') - n(A \cap B')$$

$$n(A \cup B') = 13 + 12 - 8$$

$$n(A \cup B') = 17 \quad (01 \text{ marks})$$

$$(ii) \quad n(A' \cap B) = n(B) \text{ only}$$

$$n(A' \cap B) = n(B) - n(A \cap B)$$

$$n(A' \cap B) = 8 - 5 = 3 \quad (01 \text{ marks})$$

- (c) Out of 26 animals in a zoo, 5 animals eat all types of food (grass, meat and bones), 6 animals eat grass and meat only and 4 animals eat meat and bones only, 2 animals eat grass and bones only. The number of animals eating one type of food only is divided equally between the three types of food. Using Venn diagram, find the number of animals eating grass.

$$n(G \cap M \cap B) = 5$$

$$n(G \cap M) \text{ only} = 6$$

$$n(M \cap B) \text{ only} = 4$$

$$n(G \cap B) \text{ only} = 2$$

$$\begin{aligned}
 n(M \cup B \cup G) &= n(M)_{\text{only}} + n(B)_{\text{only}} + n(G)_{\text{only}} \\
 &\quad + n(M \cap B)_{\text{only}} + n(B \cap G)_{\text{only}} + n(G \cap M)_{\text{only}} \\
 &\quad + n(M \cap B \cap G)
 \end{aligned}$$

$\left(00\frac{1}{2} \text{ marks}\right)$

$$26 = x + x + x + 4 + 5 + 6 + 2$$

$$26 = 3x + 17$$

$$3x = 26 - 17$$

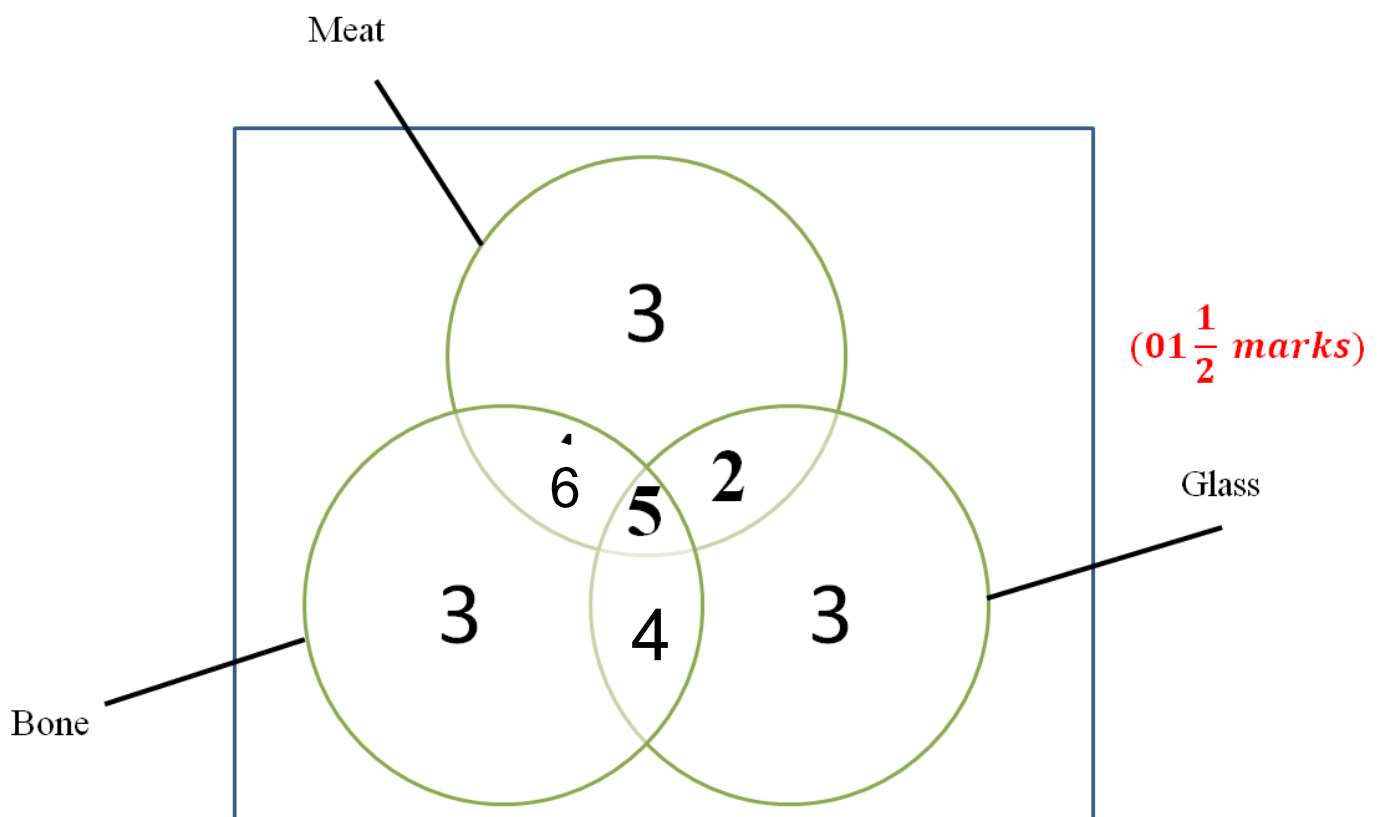
$\left(00\frac{1}{2} \text{ marks}\right)$

$$3x = 9$$

$$x = 3$$

$$n(M)_{\text{only}} = n(B)_{\text{only}} = n(G)_{\text{only}} = 3$$

$\left(00\frac{1}{2} \text{ marks}\right)$



Animal eating glass is $4 + 3 + 2 + 5 = 14$

6. (a) Given that $f(x) = 4x + 1$ and $g(x) = 6x + k$ Find the value of k if $f \circ g(x) = g \circ f(x)$

$$g(x) = 6x + k$$

$$f(x) = 4x + 1$$

$$f \circ g(x) = g \circ f(x)$$

$$4(6x + k) + 1 = 6(4x + 1) + k \quad (01mark)$$

$$\cancel{24x} + 4k + 1 = \cancel{24x} + 6 + k$$

$$4k + 1 = 6 + k$$

$$3k = 5$$

$$k = \frac{5}{3} \quad (01mark)$$

- (b) The quadratic function graph passes through the three points $(-3, 4)$

$(-2, 2)$ and $(-4, 2)$. The function in the form of $a(x - h)^2 = f(x) - k$

$$\text{Let } f(x) = ax^2 + bx + c$$

$$y = ax^2 + bx + c$$

$$\text{At } (-3, 4)$$

$$4 = 9a - 3b + c$$

$$\text{At } (-2, 2)$$

$$2 = 4a - 2b + c$$

$$\text{At } (-4, 2)$$

$$2 = 16a - 4b + c \quad (01mark)$$

Now

$$\begin{cases} 9a - 3b + c = 4 \\ 4a - 2b + c = 2 \\ 16a - 4b + c = 2 \end{cases}$$

$$a = -2, b = -12 \text{ and } c = -14 \quad (01mark)$$

$$f(x) = ax^2 + bx + c$$

$$f(x) = -2x^2 - 12x - 14$$

$$f(x) = -2(x^2 + 6x + 7)$$

$$f(x) = -2\left(x^2 + 6x + \left(\frac{6}{2}\right)^2 + 7 - \left(\frac{6}{2}\right)^2\right) \quad \left(00\frac{1}{2} \text{ marks}\right)$$

$$f(x) = -2((x+3)^2 + 7 - 9)$$

$$f(x) = -2(x+3)^2 + 4$$

$$-2(x+3)^2 = f(x) - 4 \quad \left(00\frac{1}{2} \text{ marks}\right)$$

(c) Draw the graph of the function $f(x) = \frac{x^2 - 2x + 1}{x^3 + x^2 - 2x}$

$$f(x) = \frac{x^2 - 2x + 1}{x^3 + x^2 - 2x}$$

$$f(x) = \frac{x^2 - 2x + 1}{x(x^2 + x - 2)}$$

$$f(x) = \frac{x^2 - x - x + 1}{x(x^2 + 2x - x - 2)}$$

$$f(x) = \frac{x^2 - x - x + 1}{x(x^2 - x + 2x - 2)}$$

$$f(x) = \frac{x(x-1) - 1(x-1)}{x(x-1) + 2(x-1)}$$

$$f(x) = \frac{(x-1)(x-1)}{(x+2)(x-1)} \quad \left(00\frac{1}{2} \text{ marks}\right)$$

$$x - 1 = 0 \quad x = 1$$

$$f(1) = 0 \quad \text{Hole is } (1, 0)$$

$$x - \text{intercept } y = 0$$

$$y = \frac{x-1}{x(x+2)} \Rightarrow \frac{x-1}{x(x+2)} = 0$$

$$x-1=0$$

$$x=1$$

$$x - \text{intercept is } (1, 0)$$

$\left(00\frac{1}{2} \text{ marks}\right)$

$$y - \text{intercept}$$

$$x=0$$

$$y = \frac{x-1}{x(x+2)}$$

$$y = \frac{0-1}{0(0+2)} = \alpha$$

Vertical Asymptotes

$$x(x+2)=0$$

$$x=0$$

$$x+2=0$$

$$x=-2$$

$\left(00\frac{1}{2} \text{ marks}\right)$

Horizontal Asymptotes

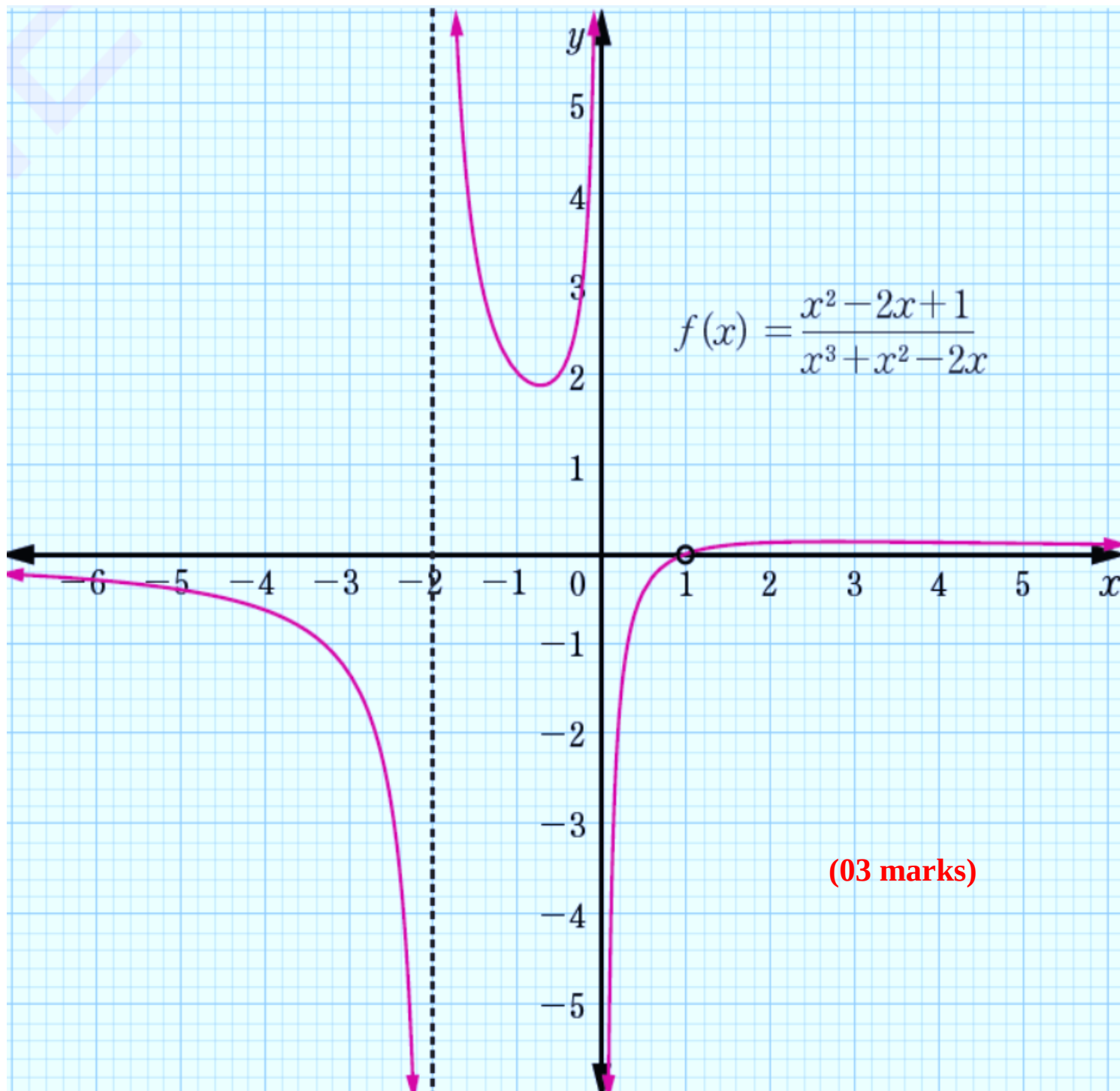
$$y = \frac{x-1}{x(x+2)}$$

$$y = \frac{x-1}{x^2+2x}$$

$$y = \frac{\frac{x}{x^2} - \frac{1}{x^2}}{\frac{x}{x^2} + \frac{2x}{x^2}} y = \frac{\frac{1}{x} - \frac{1}{x^2}}{1 + \frac{2}{x}} \quad x = \alpha$$

$$y=0$$

$\left(00\frac{1}{2} \text{ marks}\right)$



7. (a) Verify that the roots of $\cos x - xe^x = 0$ lies between 0 and 1 also by secant method to four decimal places. Perform only one iterations.

$$\cos x - xe^x = 0$$

$$f(x) = \cos x - xe^x$$

$$f(x_n) = \cos x_n - x_n e^{x_n}$$

$$x_0 = 0 \text{ and } x_1 = 1$$

$$f(x_0) = \cos x_0 - x_0 e^{x_0}$$

$$f(0) = \cos 0 - 0e^0$$

$$f(0) = 1$$

$\left(00\frac{1}{2} \text{ marks}\right)$

$$f(x_1) = \cos x_1 - x_1 e^{x_1}$$

$$f(1) = \cos 1 - e$$

$$f(1) = -2.1780$$

$$f(0) \times f(1) = -ve$$

$\left(00\frac{1}{2} \text{ marks}\right)$

Hence proved

Now use method to find root

$$x_{n+2} = \frac{x_n f(x_{n+1}) - x_{n+1} f(x_n)}{f(x_{n+1}) - f(x_n)}$$

$\left(00\frac{1}{2} \text{ marks}\right)$

$$x_2 = \frac{x_0 f(x_1) - x_1 f(x_0)}{f(x_1) - f(x_0)}$$

$$x_2 = \frac{0(-2.178) - 1(1)}{-2.1780 - 1}$$

$$x_2 = \frac{1}{3.1780} = 0.3147$$

$$x_2 = 0.3147$$

$\left(00\frac{1}{2} \text{ marks}\right)$

- (b) Estimate the value of the integral $\int_0^1 x e^{-x^2} dx$ (keep your answers in four decimal places).

(i) By Simpson's rule using five ordinates

$$h = \frac{b - a}{\text{Ordinate} - 1}$$

$$h = \frac{1 - 0}{5 - 1} = \frac{1}{4} = 0.25 \quad \left(00\frac{1}{2} \text{ marks}\right)$$

n	x	y	<i>Middle</i>	<i>1st and last</i>	<i>Odd</i>	<i>Even</i>
0	0	0		0		
1	0.25	0.2349	0.2349		0.2349	
2	0.5	0.3894	0.3894			0.3894
3	0.75	0.4273	0.4273		0.4273	
4	1	0.3679		0.3679		
Σ			1.0516	0.3679	0.6622	0.3894

$\left(01\frac{1}{2} \text{ marks}\right)$

$$\int_a^b f(x) dx = \frac{h}{3} (\text{first} + \text{last} + 2 \sum \text{Even} + 4 \sum \text{odd}) \quad \left(00\frac{1}{2} \text{ marks}\right)$$

$$\int_0^1 x e^{-x^2} dx = \frac{0.25}{3} (0.3679 + 2(0.3894) + 4(0.6622)) \quad \left(00\frac{1}{2} \text{ marks}\right)$$

$$\int_0^1 x e^{-x^2} dx = \frac{0.25}{3} (0.3679 + 3.4276) \quad \left(00\frac{1}{2} \text{ marks}\right)$$

$$\int_0^1 x e^{-x^2} dx = 0.3163 \quad \left(00\frac{1}{2} \text{ marks}\right)$$

(ii) By means of the first four terms of the Maclaurin series for e^{-x^2}

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2} - \frac{x^6}{6} \dots\dots\dots \left(00\frac{1}{2} \text{ marks}\right)$$

$$\int_0^1 xe^{-x^2} dx = \int_0^1 x \left(1 - x^2 + \frac{x^4}{2} - \frac{x^6}{6}\right) dx$$

$$\int_0^1 xe^{-x^2} dx = \int_0^1 \left(x - x^3 + \frac{x^5}{2} - \frac{x^7}{6}\right) dx \left(00\frac{1}{2} \text{ marks}\right)$$

$$\int_0^1 xe^{-x^2} dx = \left(\frac{x^2}{2} - \frac{x^4}{4} + \frac{x^6}{12} - \frac{x^8}{48}\right) \Bigg|_0^1$$

$$\int_0^1 xe^{-x^2} dx = \left(\frac{1^2}{2} - \frac{1^4}{4} + \frac{1^6}{12} - \frac{1^8}{48}\right) - 0 \left(00\frac{1}{2} \text{ marks}\right)$$

$$\int_0^1 xe^{-x^2} dx = \frac{1}{2} - \frac{1}{4} + \frac{1}{12} - \frac{1}{48}$$

$$\int_0^1 xe^{-x^2} dx = 0.3125 \left(00\frac{1}{2} \text{ marks}\right)$$

(iii) By analytical method

$$\text{Let } u = -x^2 \Rightarrow \frac{du}{dx} = -2x \Rightarrow dx = -\frac{du}{2x} \text{ Limit } u_1 = 0, u_2 = -1 \left(00\frac{1}{2} \text{ marks}\right)$$

$$\int_0^1 xe^{-x^2} dx = \int_0^{-1} xe^u \left(-\frac{du}{2x}\right)$$

$$\int_0^1 xe^{-x^2} dx = \int_0^{-1} -\frac{1}{2} e^u du \left(00\frac{1}{2} \text{ marks}\right)$$

$$\int_0^1 xe^{-x^2} dx = \frac{1}{2} \int_{-1}^0 e^u du$$

$$\int_0^1 xe^{-x^2} dx = \frac{1}{2} (e^u)_{-1}^0 \left(00\frac{1}{2} \text{ marks}\right)$$

$$\int_0^1 xe^{-x^2} dx = \frac{1}{2} (e^0 - e^{-1})$$

$$\int_0^1 xe^{-x^2} dx = 0.3161 \left(00\frac{1}{2} \text{ marks}\right)$$

8. (a) The shortest distance of the point $(p, 2)$ from the line $3x + 4y + 1 = 0$ is

$$\frac{\sqrt{2}}{5} \text{ units} \quad \text{Find the value of } p$$

$$d = \frac{\sqrt{2}}{5} \quad \text{and} \quad 3x + 4y + 1$$

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} \quad \left(00\frac{1}{2} \text{ marks}\right)$$

$$\frac{\sqrt{2}}{5} = \frac{|3x_1 + 4y_1 + 1|}{\sqrt{3^2 + 4^2}} \quad \left(00\frac{1}{2} \text{ marks}\right)$$

$$(x_1, y_1) = (p, 2)$$

$$\frac{\sqrt{2}}{5} = \frac{|3(p) + 4(2) + 1|}{\sqrt{9 + 16}} \quad \left(00\frac{1}{2} \text{ marks}\right)$$

$$\frac{\sqrt{2}}{5} = \frac{|3(p) + 8 + 1|}{\sqrt{25}} \quad \left(00\frac{1}{2} \text{ marks}\right)$$

$$\frac{\sqrt{2}}{5} = \frac{|3p + 9|}{5} \quad \left(00\frac{1}{2} \text{ marks}\right)$$

$$\sqrt{2} = |3p + 9|$$

$$\sqrt{2} = \pm(3p + 9)$$

$$p = \frac{\sqrt{2} - 9}{3} \quad \text{or} \quad p = \frac{-\sqrt{2} - 9}{3} \quad \left(00\frac{1}{2} \text{ marks}\right)$$

- (b) Find the secant of the acute angle between the pair of lines

$$y^2 + 2x^2 - 3xy + 5x - y - 12 = 0$$

$$y^2 + 2x^2 - 3xy + 5x - y - 12 = 0$$

$$y^2 - 3xy - y + 2x^2 + 5x - 12 = 0$$

$$y^2 - (3x + 1)y + 2x^2 + 5x - 12 = 0 \quad \left(00\frac{1}{2} \text{ marks}\right)$$

$$y = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$y = \frac{-(-(3x + 1)) \pm \sqrt{(-(3x + 1))^2 - 4 \times 1 \times (2x^2 + 5x - 12)}}{2 \times 1} \quad \left(00\frac{1}{2} \text{ marks}\right)$$

$$y = \frac{3x+1 \pm \sqrt{9x^2 + 6x + 1 - 8x^2 - 20x + 48}}{2}$$

$$y = \frac{3x+1 \pm \sqrt{x^2 - 14x + 49}}{2}$$

$$y = \frac{3x+1 \pm \sqrt{x^2 - 7x - 7x + 49}}{2}$$

$$y = \frac{3x+1 \pm \sqrt{x(x-7) - 7(x-7)}}{2}$$

$$y = \frac{3x+1 \pm \sqrt{(x-7)(x-7)}}{2}$$

$\left(00\frac{1}{2} \text{ marks}\right)$

$$y = \frac{3x+1 \pm (x-7)}{2}$$

$$y = \frac{3x+1+x-7}{2} \quad \text{or} \quad y = \frac{3x+1-x+7}{2}$$

$$y = \frac{4x-6}{2} \quad \text{or} \quad y = \frac{2x+8}{2}$$

$$y = 2x-3 \quad \text{or} \quad y = x+4$$

$$m_1 = 2 \quad \text{and} \quad m_2 = 1$$

$\left(00\frac{1}{2} \text{ marks}\right)$

$$\tan \theta = \frac{m_1 - m_2}{1 + m_1 m_2}$$

$$\tan \theta = \frac{2-1}{1+2 \times 1}$$

$\left(00\frac{1}{2} \text{ mark}\right)$

$$\tan \theta = \frac{1}{3} \Rightarrow \tan^2 \theta = \frac{1}{9}$$

$$1 + \tan^2 \theta = \sec^2 \theta$$

$\left(00\frac{1}{2} \text{ mark}\right)$

$$1 + \frac{1}{9} = \sec^2 \theta$$

$$\sec^2 \theta = \frac{10}{9}$$

$\left(00\frac{1}{2} \text{ mark}\right)$

$$\sec \theta = \sqrt{\frac{10}{9}} = \frac{\sqrt{10}}{3}$$

$$\sec \theta = \sqrt{\frac{10}{9}} = \frac{\sqrt{10}}{3}$$

$\left(00\frac{1}{2} \text{ mark}\right)$

- (c) A bicycle tire has equation $x^2 + y^2 - 4x - 6y + 11 = 0$ find the coordinate of the hub and approximately length of spokes.

$$x^2 + y^2 - 4x - 6y + 11 = 0$$

Compare

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

$$2gx = -4x$$

$$g = -2$$

$$2fy = -6y$$

$$f = -3$$

$$c = 11$$

Centre is (2, 3)

$\left(01\frac{1}{2} \text{ mark}\right)$

$$r = \sqrt{g^2 + f^2 - c}$$

$$r = \sqrt{2^2 + 3^2 - 11}$$

$$r = \sqrt{4 + 9 - 11}$$

$$r = \sqrt{2} \text{ units}$$

$\left(01\frac{1}{2} \text{ mark}\right)$

9. (a) Prove that derivative of $\sqrt{\frac{1+\sin x}{1-\sin x}}$ is $\frac{1}{1-\sin x}$

$$y = \sqrt{\frac{1+\sin x}{1-\sin x}}$$

$$y = \sqrt{\frac{(1+\sin x)(1+\sin x)}{(1-\sin x)(1+\sin x)}} \quad \left(00\frac{1}{2} \text{ mark}\right)$$

$$y = \sqrt{\frac{(1+\sin x)^2}{1-\sin^2 x}}$$

$$y = \sqrt{\frac{(1+\sin x)^2}{\cos^2 x}} \quad \left(00\frac{1}{2} \text{ mark}\right)$$

$$y = \sqrt{\left(\frac{1+\sin x}{\cos x}\right)^2}$$

$$y = \frac{1+\sin x}{\cos x} \quad \left(00\frac{1}{2} \text{ mark}\right)$$

$$y = \sec x + \tan x$$

$$\frac{dy}{dx} = \sec x \tan x + \sec^2 x$$

$$\frac{dy}{dx} = \frac{1}{\cos x} \frac{\sin x}{\cos x} + \frac{1}{\cos^2 x} \quad \left(00\frac{1}{2} \text{ mark}\right)$$

$$\frac{dy}{dx} = \frac{\sin x}{\cos^2 x} + \frac{1}{\cos^2 x} \quad \left(00\frac{1}{2} \text{ mark}\right)$$

$$\frac{dy}{dx} = \frac{\sin x + 1}{\cos^2 x}$$

$$\frac{dy}{dx} = \frac{1+\sin x}{1-\sin^2 x} \quad \left(00\frac{1}{2} \text{ mark}\right)$$

$$\frac{dy}{dx} = \frac{\cancel{1+\sin x}}{(1-\sin x)(\cancel{1+\sin x})} \quad \left(00\frac{1}{2} \text{ mark}\right)$$

$$\frac{dy}{dx} = \frac{1}{1-\sin x} \quad \left(00\frac{1}{2} \text{ mark}\right)$$

- (b) Find the derivative of $y = (\tan x + \sec x)^n$ where n is positive integer

$$y = (\tan x + \sec x)^n$$

Let $u = \tan x + \sec x$

$$\frac{du}{dx} = \sec^2 x + \tan x \sec x \quad \left(00\frac{1}{2} \text{ marks} \right)$$

$$y = u^n$$

$$\frac{dy}{du} = nu^{n-1}$$

$$\frac{dy}{du} = n(\tan x + \sec x)^{n-1} \quad \left(00\frac{1}{2} \text{ marks} \right)$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

$$\frac{dy}{dx} = n(\tan x + \sec x)^{n-1} \times (\sec^2 x + \tan x \sec x)$$

$$\frac{dy}{dx} = n(\sec^2 x + \tan x \sec x)(\tan x + \sec x)^{n-1}$$

$$\frac{dy}{dx} = n \sec x (\sec x + \tan x)(\tan x + \sec x)^{n-1} \quad \left(00\frac{1}{2} \text{ marks} \right)$$

$$\frac{dy}{dx} = n \sec x (\tan x + \sec x)(\tan x + \sec x)^{n-1}$$

$$\frac{dy}{dx} = n \sec x (\tan x + \sec x)^n \quad \left(00\frac{1}{2} \text{ marks} \right)$$

- (c) A chemical factory wishes to make a cylindrical container of thin metal to hold 126 cm^3 , using the least possible area of a metal. If the outside surface area $A \text{ cm}^2$ and the radius is $r \text{ cm}$, then

- (i) Show that

$$A = \frac{2}{r}(\pi r^3 + 126)$$

$$v = \pi r^2 h$$

$$126 = \pi r^2 h$$

$$h = \frac{126}{\pi r^2}$$

$$A = 2\pi r^2 + 2\pi r h \quad \left(00\frac{1}{2} \text{ marks} \right)$$

$$A = 2\pi r^2 + 2\pi r \left(\frac{126}{\pi r^2} \right)$$

$$A = 2\pi r^2 + \frac{256 \cancel{\pi r}}{\cancel{\pi r} r}$$

$$A = 2\pi r^2 + \frac{256}{r}$$

$$A = \frac{2}{r} (\pi r^3 + 126) \quad \left(00 \frac{1}{2} \text{ marks} \right)$$

(iii) Find the required radius and height of the container.

$$A = \frac{2}{r} (\pi r^3 + 126)$$

$$A = 2\pi r^2 + \frac{252}{r} \quad \left(00 \frac{1}{2} \text{ marks} \right)$$

$$\frac{dA}{dr} = 4\pi r - \frac{252}{r^2}$$

$$\frac{d^2A}{dr^2} = 4\pi + \frac{504}{r^3} \quad \left(00 \frac{1}{2} \text{ marks} \right)$$

On a minimum point

$$4\pi r - \frac{252}{r^2} = 0$$

$$4\pi r = \frac{252}{r^2} \quad \left(00 \frac{1}{2} \text{ marks} \right)$$

$$4\pi r^3 = 252$$

$$r^3 = \frac{252}{4\pi} = \frac{63}{\pi} \quad \left(00 \frac{1}{2} \text{ marks} \right)$$

$$r = \sqrt[3]{\frac{63}{\pi}} = 2.72 \text{ cm} \quad \left(00 \frac{1}{2} \text{ marks} \right)$$

$$h = \frac{126}{\pi r^2}$$

$$h = \frac{126}{\pi (2.72)^2} = 5.43 \text{ cm} \quad \left(00 \frac{1}{2} \text{ marks} \right)$$

10. (a) Evaluate $\int \frac{x^2}{\sqrt{4-x^2}}$

$$\int \frac{x^2}{\sqrt{4-x^2}}$$

$$\text{Let } x = 2 \sin \theta \Rightarrow \theta = \sin^{-1} \frac{x}{2}$$

$$\frac{dx}{d\theta} = 2 \cos \theta \quad \left(00 \frac{1}{2} \text{ marks} \right)$$

$$dx = 2 \cos \theta d\theta$$

$$= \int \frac{(2 \sin \theta)^2}{\sqrt{4 - (2 \sin \theta)^2}} 2 \cos \theta d\theta \quad \left(00 \frac{1}{2} \text{ marks} \right)$$

$$= \int \frac{4 \sin^2 \theta}{\sqrt{4 - 4 \sin^2 \theta}} 2 \cos \theta d\theta \quad \left(00 \frac{1}{2} \text{ marks} \right)$$

$$= \int \frac{8 \sin^2 \theta}{2 \sqrt{1 - \sin^2 \theta}} \cos \theta d\theta \quad \left(00 \frac{1}{2} \text{ marks} \right)$$

$$= \int \frac{4 \sin^2 \theta}{\cancel{\cos \theta}} \cancel{\cos \theta} d\theta \quad \left(00 \frac{1}{2} \text{ marks} \right)$$

$$= \int 4 \sin^2 \theta d\theta$$

$$= \int 4 \left(\frac{1 - \cos 2\theta}{2} \right) d\theta \quad \left(00 \frac{1}{2} \text{ marks} \right)$$

$$= \int 2 d\theta - \int 2 \cos 2\theta d\theta$$

$$= 2\theta - \sin 2\theta + C$$

$$\int \frac{x^2}{\sqrt{4-x^2}} = 2 \sin^{-1} \left(\frac{x}{2} \right) - \sin 2 \left(\sin^{-1} \left(\frac{x}{2} \right) \right) + C \quad \left(00 \frac{1}{2} \text{ marks} \right)$$

more simplification

$$\int \frac{x^2}{\sqrt{4-x^2}} = 2 \sin^{-1} \left(\frac{x}{2} \right) - 2 \cos \left(\sin^{-1} \left(\frac{x}{2} \right) \right) \sin \left(\sin^{-1} \left(\frac{x}{2} \right) \right) + C$$

$$\int \frac{x^2}{\sqrt{4-x^2}} = 2 \sin^{-1} \left(\frac{x}{2} \right) - x \sqrt{1 - \left(\frac{x}{2} \right)^2} + C$$

$$\int \frac{x^2}{\sqrt{4-x^2}} = 2 \sin^{-1} \left(\frac{x}{2} \right) - \frac{x}{2} \sqrt{4-x^2} + C \quad \left(00 \frac{1}{2} \text{ marks} \right)$$

(b) Show that $\int_0^1 \frac{x^2}{5-x^2} dx = -\frac{11}{2} + 25 \ln\left(\frac{5}{4}\right)$

$$\int_0^1 \frac{x^2}{5-x^2} dx$$

$$5-x \Bigg) \frac{-x-5}{x^2}$$

$$\frac{x^2-5x}{5x}$$

$$5x$$

$$\underline{5x-25}$$

$$25$$

$$\left(00\frac{1}{2} \text{ marks}\right)$$

$$\int_0^1 \frac{x^2}{5-x^2} dx = \int_0^1 \left(-5-x + \frac{25}{5-x}\right) dx$$

$$\int_0^1 \frac{x^2}{5-x^2} dx = \int_0^1 (-5-x) dx + \int_0^1 \frac{25}{5-x} dx$$

$$\left(00\frac{1}{2} \text{ marks}\right)$$

$$\int_0^1 \frac{x^2}{5-x^2} dx = \left[-5x - \frac{x^2}{2}\right]_0^1 + 25 \left[-\ln(5-x)\right]_0^1$$

$$\int_0^1 \frac{x^2}{5-x^2} dx = \left[-5x - \frac{x^2}{2}\right]_0^1 - 25 \left[\ln(5-x)\right]_0^1$$

$$\left(00\frac{1}{2} \text{ marks}\right)$$

$$\int_0^1 \frac{x^2}{5-x^2} dx = \left(-5 - \frac{1}{2}\right) - 25 \left[\ln(5-x)\right]_0^1$$

$$\int_0^1 \frac{x^2}{5-x^2} dx = -\frac{11}{2} - 25 \left[\ln(5-1) - \ln(5-0)\right]$$

$$\left(00\frac{1}{2} \text{ marks}\right)$$

$$\int_0^1 \frac{x^2}{5-x^2} dx = -\frac{11}{2} - 25 \left[\ln 4 - \ln 5\right]$$

$$\left(00\frac{1}{2} \text{ marks}\right)$$

$$\int_0^1 \frac{x^2}{5-x^2} dx = -\frac{11}{2} + 25 \left[\ln 5 - \ln 4\right]$$

$$\int_0^1 \frac{x^2}{5-x^2} dx = -\frac{11}{2} + 25 \ln\left(\frac{5}{4}\right)$$

$$\left(00\frac{1}{2} \text{ marks}\right)$$

(c) Find the length of the curve given by the parametric equations

$$x = 3 \sin 3\theta \quad \text{and} \quad y = 3 \cos 3\theta, \quad 0 \leq \theta \leq 2\pi$$

$$x = 3 \sin 3\theta$$

$$y = 3 \cos 3\theta$$

$$\frac{dx}{d\theta} = 9 \cos 3\theta$$

$\left(00\frac{1}{2} \text{ marks} \right)$

$$\frac{dy}{d\theta} = -9 \sin 3\theta$$

$\left(00\frac{1}{2} \text{ marks} \right)$

$$l = \int_{\theta_1}^{\theta_2} \sqrt{\left(\frac{dy}{d\theta}\right)^2 + \left(\frac{dx}{d\theta}\right)^2} d\theta$$

$\left(00\frac{1}{2} \text{ marks} \right)$

$$l = \int_0^{2\pi} \sqrt{(-9 \sin 3\theta)^2 + (9 \cos 3\theta)^2} d\theta$$

$\left(00\frac{1}{2} \text{ marks} \right)$

$$l = \int_0^{2\pi} \sqrt{81 \sin^2 3\theta + 81 \cos^2 3\theta} d\theta$$

$$l = \int_0^{2\pi} \sqrt{81(\sin^2 3\theta + \cos^2 3\theta)} d\theta$$

$\left(00\frac{1}{2} \text{ marks} \right)$

$$l = \int_0^{2\pi} 9 d\theta$$

$$l = [9\theta]_0^{2\pi}$$

$$l = 9(2\pi - 0)$$

$$l = 18\pi \text{ units}$$

$\left(00\frac{1}{2} \text{ marks} \right)$